

SUBJECT : PROBLEM SOLVING AND TESTING USING JAVA

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Task 2 (Difficult): Real-Time Stream Analytics Engine

Problem Description

A cloud analytics platform receives millions of sensor readings. Each reading contains a sensor ID and temperature value. Using stream processing concepts, perform the following operations:

1. Filter temperatures greater than 50.
2. Group readings by sensor ID.
3. Compute average temperature per sensor.
4. Sort sensors based on average temperature in descending order.

Input Format

First line contains integer N.

Next N lines contain SensorID and Temperature.

Output Format

Display SensorID and average temperature sorted in descending order.

Constraints

$$1 \leq N \leq 10^5$$

$$0 \leq \text{Temperature} \leq 100$$

Sample Input

```
6
S1 60
S2 40
S1 80
S3 70
```

S2 90

S3 30

Sample Output

S1 70.0

S2 90.0

S3 70.0

PROGRAM :

```
import java.util.*;
```

```
import java.util.stream.*;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        Map<String, List<Double>> data = new HashMap<>();
```

```
        for(int i = 0; i < n; i++) {
```

```
            String sensor = sc.next();
```

```
            double temp = sc.nextDouble();
```

```
            //Filter temperature > 50
```

```
            if(temp > 50) {
```

```

        data.computeIfAbsent(sensor, k -> new
ArrayList<>()).add(temp);
    }
}

//Calculate average and sort in descending order
data.entrySet().stream()
    .map(e->new AbstractMap.SimpleEntry<>(
        e.getKey(),
        e.getValue().stream()
            .mapToDouble(Double::doubleValue)
            .average()
            .orElse(0)
    ))
    .sorted((a,b) -> Double.compare(b.getValue(), a.getValue()))
    .forEach(e-> System.out.println(
        e.getKey() + " " + e.getValue()
    ));

sc.close();
}
}

```

OUTPUT:

```
● apple@Nandinis-Macbook-Air Java % javac Lambda.java
● apple@Nandinis-Macbook-Air Java % java Lambda.java
5
S1 40
S2 70
S3 30
S4 20
S5 50
S2 70.0
○ apple@Nandinis-Macbook-Air Java %
```